



*In this chapter you will learn about...*

- ☐ Triangle, types of triangles
- ☐ Angle sum property of a triangle
- ☐ Exterior and interior opposite angles of a triangle
- ☐ Medians of a triangle
- ☐ Altitudes of a triangle
- ☐ Triangle inequality property
- ☐ Pythagoras Theorem

## INTRODUCTION

We have learnt about lines, rays, line segments and angles in the previous chapter. Now, we will study about a plane figure formed by three non-parallel line segments called a triangle.

## TRIANGLE

A closed figure formed by three line segments is called a **triangle**.

Consider the adjoining figure.

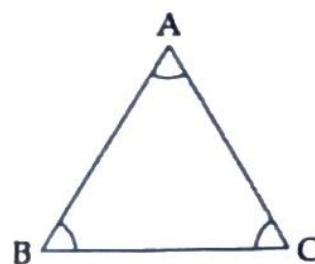
A triangle has

- (i) three sides –  $\overline{AB}$ ,  $\overline{BC}$  and  $\overline{CA}$
- (ii) three angles –  $\angle A$ ,  $\angle B$  and  $\angle C$
- (iii) three vertices – A, B and C.

A triangle with vertices A, B and C is denoted by  $\triangle ABC$ .

In the given figure, side BC is opposite to the vertex A, side AC is opposite to the vertex B and side AB is opposite to the vertex C.

The three sides and three angles of a triangle are called the six parts or elements of a triangle.



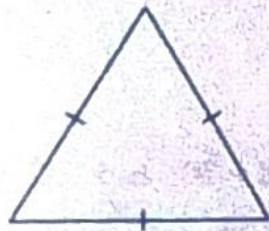
## TYPES OF TRIANGLES

Triangles can be classified according to the lengths of their sides and according to their interior angles.

### Classification on the Basis of Sides

- (i) A triangle in which all the three sides are equal is called an **equilateral triangle**.
- (ii) A triangle in which any two of its sides are of equal lengths is called an **isosceles triangle**.

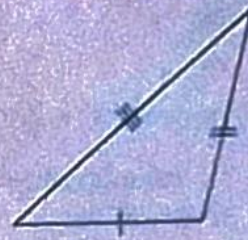
(iii) A triangle in which all three sides are of different lengths is called a **scalene triangle**.



Equilateral triangle



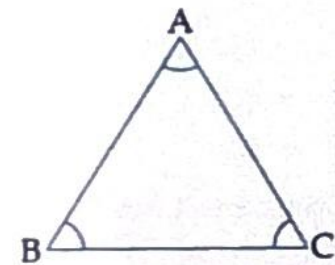
Isosceles triangle



Scalene triangle

### Classification on the Basis of Angles

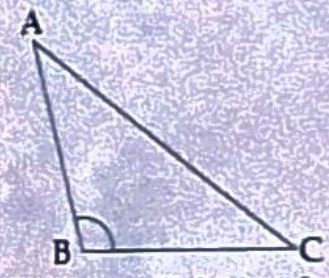
- (i) A triangle in which all the angles are less than  $90^\circ$  (acute) is called an **acute angled triangle**.
- (ii) A triangle having its one angle as a right angle is called a **right angled triangle**.
- (iii) A triangle having one of its angles more than  $90^\circ$  (obtuse) is called an **obtuse angled triangle**.



Acute angled triangle



Right angled triangle



Obtuse angled triangle

### ANGLE SUM PROPERTY OF A TRIANGLE

The sum of the angles of a triangle is  $180^\circ$ .

**Proof :** Let ABC be any triangle. Through A, draw a line PQ parallel to the side BC of the triangle.

Since  $PQ \parallel BC$  and AB is a transversal.

$$\therefore \angle ABC = \angle PAB \quad \dots(i) \quad (\text{Alternate angles are equal})$$

Similarly,  $PQ \parallel BC$  and AC is a transversal.

$$\therefore \angle ACB = \angle QAC \quad \dots(ii) \quad (\text{Alternate angles are equal})$$

Adding equations (i) and (ii), we get

$$\angle ABC + \angle ACB = \angle PAB + \angle QAC$$

Adding  $\angle BAC$  on both sides, we get

$$\angle ABC + \angle ACB + \angle BAC = \angle PAB + \angle QAC + \angle BAC$$

$$\Rightarrow \angle ABC + \angle ACB + \angle BAC = 180^\circ \quad (\text{Linearity property})$$

Hence, the sum of the angles of a triangle is  $180^\circ$  i.e., 2 right angles.

The following examples illustrate the application of this property.

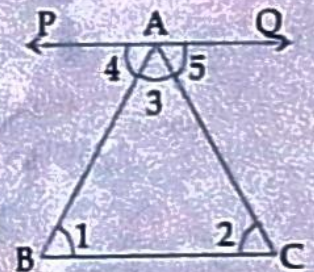
**Example 1 :** In a  $\triangle ABC$ , if  $\angle A = 70^\circ$ ,  $\angle B = 65^\circ$ , find the measure of  $\angle C$ .

**Solution :** We have,  $\angle A = 70^\circ$ ,  $\angle B = 65^\circ$

$\therefore$  Sum of the angles of a triangle is  $180^\circ$ .

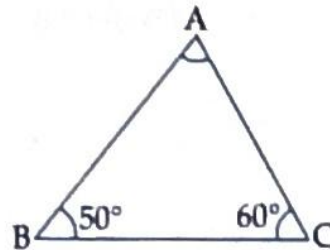
$$\therefore \angle A + \angle B + \angle C = 180^\circ$$

$$\Rightarrow 70^\circ + 65^\circ + \angle C = 180^\circ$$



$$\begin{aligned} \Rightarrow 135^\circ + \angle C &= 180^\circ \\ \Rightarrow \angle C &= 180^\circ - 135^\circ \\ \Rightarrow \angle C &= 45^\circ \\ \text{Hence, } \angle C &= 45^\circ. \end{aligned}$$

**Example 2 :** In the adjoining figure, ABC is a triangle with  $\angle B = 50^\circ$  and  $\angle C = 60^\circ$ , find the measure of  $\angle A$ .



**Solution :** In  $\triangle ABC$ ,

$$\begin{aligned} \angle A + \angle B + \angle C &= 180^\circ && \text{(Angle sum property)} \\ \Rightarrow \angle A + 50^\circ + 60^\circ &= 180^\circ && \text{(given)} \\ \Rightarrow \angle A + 110^\circ &= 180^\circ \\ \Rightarrow \angle A &= 180^\circ - 110^\circ \\ \Rightarrow \angle A &= 70^\circ \\ \text{Hence, } \angle A &= 70^\circ. \end{aligned}$$

**Example 3 :** One of the angles of a triangle is  $70^\circ$  and the other two angles are equal. Find these two angles.

**Solution :** Let  $\angle A$ ,  $\angle B$  and  $\angle C$  be three angles of a triangle.

Let  $\angle A = 70^\circ$  and  $\angle B = \angle C$ .

$$\begin{aligned} \text{We know that } \angle A + \angle B + \angle C &= 180^\circ && \text{(Angle sum property)} \\ \Rightarrow 70^\circ + \angle B + \angle B &= 180^\circ && (\because \angle C = \angle B) \\ \Rightarrow 70^\circ + 2\angle B &= 180^\circ \\ \Rightarrow 2\angle B &= 180^\circ - 70^\circ = 110^\circ \end{aligned}$$

$$\text{Hence, } \angle B = \frac{110^\circ}{2} = 55^\circ$$

Hence, the measure of each of the remaining two angles is  $55^\circ$ .

**Example 4 :** The angles of a triangle are in the ratio 1 : 2 : 3. Determine the angles.

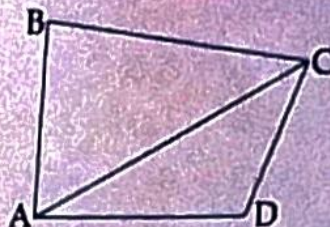
**Solution :** Let the three angles of the triangle be  $x$ ,  $2x$  and  $3x$ .

$$\begin{aligned} \text{Then, } x + 2x + 3x &= 180^\circ && \text{(Angle sum property)} \\ \Rightarrow 6x &= 180^\circ \\ \Rightarrow x &= 30^\circ \\ \therefore 2x &= 60^\circ, 3x = 90^\circ \end{aligned}$$

Hence, the required angles are  $30^\circ$ ,  $60^\circ$  and  $90^\circ$ .

**Example 5 :** The given figure is made up of two triangles.

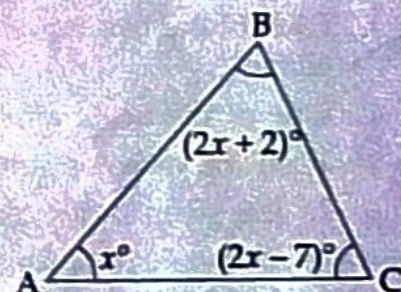
Prove that  $\angle A + \angle B + \angle C + \angle D = 360^\circ$ .



**Solution :** In  $\triangle ABC$ ,  
 $\angle CAB + \angle ABC + \angle BCA = 180^\circ$  ... (i) (Angle sum property)  
 In  $\triangle ADC$ ,  
 $\angle CAD + \angle ADC + \angle DCA = 180^\circ$  ... (ii) (Angle sum property)  
 Adding (i) and (ii), we get  
 $\angle CAB + \angle ABC + \angle BCA + \angle CAD + \angle ADC + \angle DCA = 180^\circ + 180^\circ$   
 $\Rightarrow (\angle CAB + \angle CAD) + (\angle ABC + \angle ADC) + (\angle BCA + \angle DCA) = 360^\circ$  ... (iii)  
 But  $\angle CAB + \angle CAD = \angle BAD$   
 and  $\angle BCA + \angle DCA = \angle DCB$   
 $\therefore \angle BAD + \angle ABC + \angle ADC + \angle DCB = 360^\circ$   
 or  $\angle A + \angle B + \angle D + \angle C = 360^\circ$   
 $\Rightarrow \angle A + \angle B + \angle C + \angle D = 360^\circ$

**Example 6 :** From the adjoining figure, find the value of  $x$  and also the measures of  $\angle A$ ,  $\angle B$  and  $\angle C$ .

**Solution :** In  $\triangle ABC$ ,  
 $\angle A = x^\circ$ ,  $\angle B = (2x + 2)^\circ$  and  $\angle C = (2x - 7)^\circ$   
 Since, sum of the angles of a triangle is  $180^\circ$ .  
 $\therefore \angle A + \angle B + \angle C = 180^\circ$   
 $\Rightarrow x^\circ + (2x + 2)^\circ + (2x - 7)^\circ = 180^\circ$   
 $\Rightarrow 5x - 5 = 180$   
 $\Rightarrow 5x = 180 + 5$   
 $\Rightarrow 5x = 185$   
 $\Rightarrow x = 37$

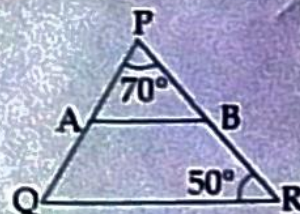


$\angle A = x^\circ = 37^\circ$ ,  $\angle B = (2x + 2)^\circ = 76^\circ$ ,  $\angle C = (2x - 7)^\circ = 67^\circ$ .

Hence,  $x = 37$  and the measures of  $\angle A$ ,  $\angle B$  and  $\angle C$  are  $37^\circ$ ,  $76^\circ$  and  $67^\circ$  respectively.

**Example 7 :** In the given  $\triangle PQR$ ,  $AB \parallel QR$ . If  $\angle P = 70^\circ$ ,  $\angle R = 50^\circ$ , find

- (i)  $\angle PBA$
- (ii)  $\angle PAB$
- (iii)  $\angle PQR$



**Solution :** (i) Since  $AB \parallel QR$  and  $PR$  is a transversal.  
 Then,  $\angle PBA = \angle BRQ$  (Corresponding angles are equal)  
 $\therefore \angle PBA = 50^\circ$

(ii) In  $\triangle PAB$ ,  
 $\angle PBA + \angle APB + \angle PAB = 180^\circ$  (Angle sum property)

$$\Rightarrow 50^\circ + 70^\circ + \angle PAB = 180^\circ$$

$$\Rightarrow 120^\circ + \angle PAB = 180^\circ$$

$$\Rightarrow \angle PAB = 180^\circ - 120^\circ$$

$$\angle PAB = 60^\circ$$

Hence,  $\angle PAB = 60^\circ$ .

(iii) In  $\triangle PQR$ ,

$$\angle QPR + \angle PRQ + \angle PQR = 180^\circ$$

(Angle sum property)

$$\Rightarrow 70^\circ + 50^\circ + \angle PQR = 180^\circ$$

$$\Rightarrow 120^\circ + \angle PQR = 180^\circ$$

$$\therefore \angle PQR = 180^\circ - 120^\circ = 60^\circ$$

Hence,  $\angle PQR = 60^\circ$ .



## EXERCISE 12.1

1. Which of the following triplets can be angles of a triangle?

(i)  $76^\circ, 64^\circ, 40^\circ$

(ii)  $30^\circ, 70^\circ, 90^\circ$

(iii)  $56^\circ, 62^\circ, 62^\circ$

(iv)  $20^\circ, 148^\circ, 30^\circ$

2. Two angles of a triangle are of measures  $60^\circ$  and  $75^\circ$ . Find the measure of the third angle.

3. Each of the two equal angles of a triangle is twice the third angle. Find the angles of the triangle.

4. One of the acute angles of a right angled triangle is  $42^\circ$ . Find the other acute angle.

5. If all the three angles of a triangle are equal then find the measure of each angle.

6. In the given figure,  $\triangle ACB$  is a right angled triangle, right angled at  $C$  and  $CD \perp AB$ .

- If  $\angle B = 40^\circ$ , then find :

(i)  $\angle BAC$

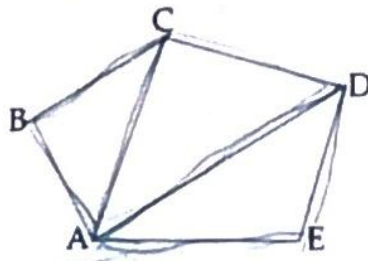
(ii)  $\angle ACD$

(iii)  $\angle BCD$



7. Two acute angles of a right angled triangle are equal. Find the two angles.

8. In the given five-sided closed figure, find  $\angle EAB + \angle ABC + \angle BCD + \angle CDE + \angle DEA$ .



9. The angles of a triangle are in the ratio  $5 : 4 : 3$ . Find the measure of each angle.

10. In  $\triangle ABC$ ,  $2\angle A = 3\angle B = 6\angle C$ . Calculate  $\angle A$ ,  $\angle B$  and  $\angle C$ .

[Hint :  $2\angle A = 3\angle B \Rightarrow \angle A : \angle B = 3 : 2$ , and  $3\angle B = 6\angle C \Rightarrow \angle B : \angle C = 6 : 3 = 2 : 1$ ]

11. The angles of a triangle are  $(x + 21)^\circ$ ,  $x^\circ$  and  $(2x - 45)^\circ$ . Find the value of  $x$ .

## EXTERIOR AND INTERIOR OPPOSITE ANGLES OF A TRIANGLE

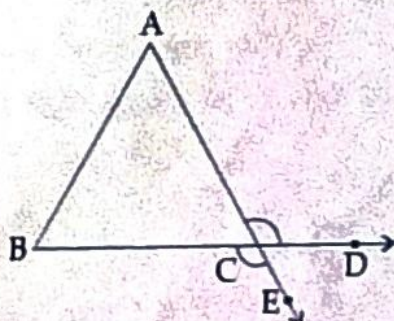
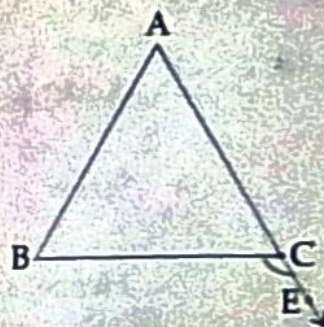
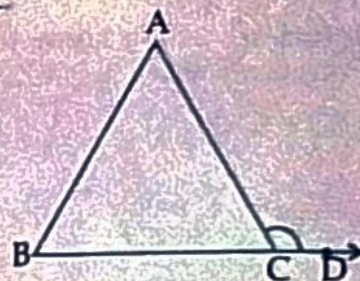
Let the side BC of a  $\triangle ABC$  be produced to D.

- (i) **Exterior angle** :  $\angle ACD$  is called an exterior angle of  $\triangle ABC$  at C.
- (ii) **Interior opposite angles** : With respect to exterior  $\angle ACD$  of  $\triangle ABC$  at C,  $\angle ABC$  and  $\angle CAB$  are called interior opposite angles.
- (iii) **Interior adjacent angle** : With respect to exterior  $\angle ACD$  of  $\triangle ABC$  at C,  $\angle ACB$  is called the interior adjacent angle.

Similarly, if the side AC of  $\triangle ABC$  is produced to E, then

- (i) **Exterior angle** :  $\angle BCE$  is called an exterior angle of  $\triangle ABC$  at C.
- (ii) **Interior opposite angles** : With respect to exterior  $\angle BCE$  of  $\triangle ABC$  at C,  $\angle BAC$  and  $\angle ABC$  are called the interior opposite angles.
- (iii) **Interior adjacent angle** : With respect to exterior  $\angle BCE$  at C,  $\angle ACB$  is called the interior adjacent angle.

Thus, at each vertex of a triangle, we have a pair of exterior angles of the triangle which are equal to each other being vertically opposite angles.



**Theorem :** Exterior angle property of a triangle : If a side of a triangle is produced, then the exterior angle so formed is equal to sum of two interior opposite angles.

**Given :**  $\triangle ABC$

**Construction :** Extend side BC to D.

**To prove that :**  $\angle 4 = \angle 1 + \angle 2$

**Proof :** In  $\triangle ABC$ ,

$$\angle 1 + \angle 2 + \angle 3 = 180^\circ \dots (i) \text{ (angle sum property)}$$

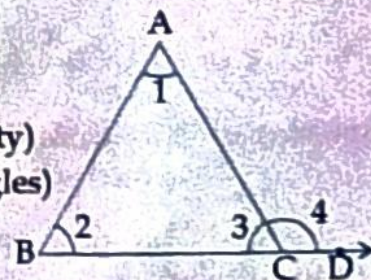
$$\text{Also, } \angle 3 + \angle 4 = 180^\circ \dots (ii) \text{ (linear pair of angles)}$$

From (i) and (ii), we get

$$\angle 3 + \angle 4 = \angle 1 + \angle 2 + \angle 3$$

$$\Rightarrow \angle 4 = \angle 1 + \angle 2$$

Hence, an exterior angle of a triangle is equal to the sum of interior opposite angles.



**Important Result :** Since  $\angle 4 = \angle 1 + \angle 2$

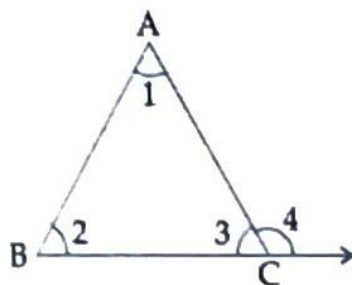
$$\Rightarrow \angle 4 > \angle 1 \text{ and } \angle 4 > \angle 2$$

Thus, in a triangle an exterior angle is greater than either of the interior opposite angles.

The following examples illustrate the application of this theorem.

**Example 1 :** By using theorem of exterior and interior opposite angles, prove that the sum of the angles of a triangle is  $180^\circ$ .

**Solution :** We know that,  $\angle 4 = \angle 1 + \angle 2$  (Exterior angle property)



Adding  $\angle 3$  on both sides, we get

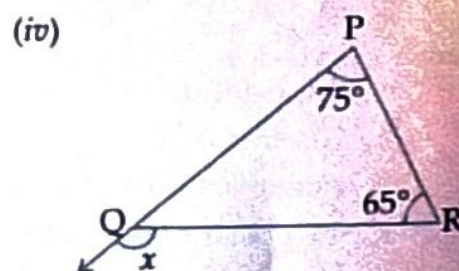
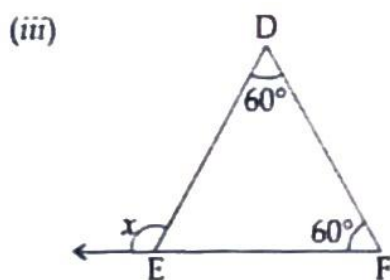
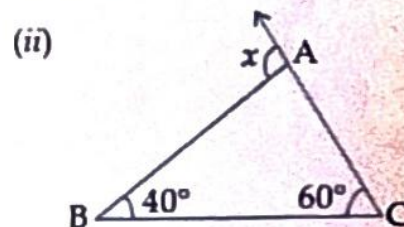
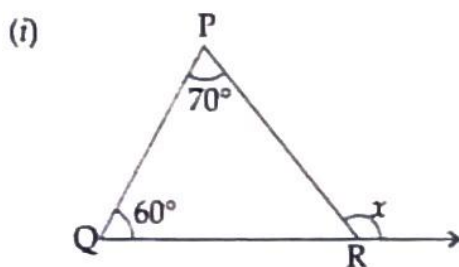
$$\angle 4 + \angle 3 = \angle 1 + \angle 2 + \angle 3$$

Since  $\angle 4 + \angle 3 = 180^\circ$  (Linear pair of angles)

$$\therefore \angle 1 + \angle 2 + \angle 3 = 180^\circ$$

Hence, sum of the angles of a triangle is  $180^\circ$ .

**Example 2 :** Find the value of  $x$  in each of the following figures :



**Solution :** (i) In  $\Delta PQR$ ,  $\angle x = 60^\circ + 70^\circ$   
 $= 130^\circ$

(ii) In  $\Delta ABC$ ,  $\angle x = 40^\circ + 60^\circ$   
 $= 100^\circ$

(iii) In  $\Delta DEF$ ,  $\angle x = 60^\circ + 60^\circ$   
 $= 120^\circ$

(iv) In  $\Delta PQR$ ,  $\angle x = 75^\circ + 65^\circ$   
 $= 140^\circ$

(Exterior angle property)

(Exterior angle property)

(Exterior angle property)

(Exterior angle property)

**Example 3 :** An exterior angle of a triangle is of measure  $80^\circ$  and one of its interior opposite angles is of measure  $25^\circ$ . Find the measure of the other interior opposite angle.

**Solution :** Let  $\angle x$  and  $\angle y$  be interior opposite angles with respect to exterior angle  $80^\circ$ .

Let  $\angle x = 25^\circ$

$\therefore 80^\circ = \angle x + \angle y$  (Exterior angle property)

$\Rightarrow 80^\circ = 25^\circ + \angle y$  ( $\angle x = 25^\circ$ )

$\Rightarrow \angle y = 80^\circ - 25^\circ$

$\Rightarrow \angle y = 55^\circ$

Hence, the other interior opposite angle is  $55^\circ$ .

**Example 4 :** The two interior opposite angles of a triangle are  $60^\circ$  and  $80^\circ$ . Find the measure of the exterior angle.

**Solution :**

$$\begin{aligned}\text{Exterior angle} &= \text{Sum of the interior opposite angles} \\ &= 60^\circ + 80^\circ \\ &= 140^\circ\end{aligned}$$

Hence, the measure of exterior angle is  $140^\circ$ .

**Example 5 :** One of the exterior angles of a triangle is  $70^\circ$  and the interior opposite angles are in the ratio 3 : 4. Find the angles of the triangle.

**Solution :** Exterior angle of the triangle =  $70^\circ$

Let the interior opposite angles be  $3x$  and  $4x$ .

$\therefore 3x + 4x = 70^\circ$  (Exterior angle property)

or  $7x = 70^\circ$

$\Rightarrow x = 10^\circ$

$\therefore 3x = 3 \times 10^\circ = 30^\circ, 4x = 4 \times 10^\circ = 40^\circ$

Since the sum of the angles of a triangle is  $180^\circ$ .

$\therefore 30^\circ + 40^\circ + \text{third angle} = 180^\circ$

$\Rightarrow 70^\circ + \text{third angle} = 180^\circ$

$\Rightarrow \text{third angle} = 180^\circ - 70^\circ = 110^\circ$

Hence, the measures of the angles of the triangle are  $30^\circ, 40^\circ$  and  $110^\circ$ .

**Example 6 :** An exterior angle of a triangle measures  $130^\circ$  and its interior opposite angles are equal to each other. Find the measure of each angle of the triangle.

**Solution :** Let each interior opposite angle be  $x$ .

Then,  $130^\circ = x + x$  (Exterior angle property)

$\Rightarrow 130^\circ = 2x$

$\Rightarrow x = 65^\circ$

Hence, measure of angles are  $65^\circ, 65^\circ$  and  $(180^\circ - 130^\circ) = 50^\circ$ . (Angle sum property)

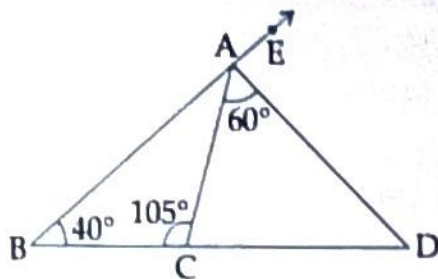
**Example 7 :** In the adjoining figure, side BA of  $\triangle ABC$  is extended to E.

Find the measure of :

(i)  $\angle ACD$

(ii)  $\angle ADC$

(iii)  $\angle DAE$



**Solution :** (i) In  $\triangle ABD$ ,  $\angle ACB$  and  $\angle ACD$  form a linear pair.

$$\therefore \angle ACB + \angle ACD = 180^\circ$$

$$\Rightarrow 105^\circ + \angle ACD = 180^\circ$$

$$\Rightarrow \angle ACD = 180^\circ - 105^\circ$$

$$\Rightarrow \angle ACD = 75^\circ$$

Hence,  $\angle ACD = 75^\circ$ .

(ii) In  $\triangle ACD$ ,

$$\angle ACB = \angle CAD + \angle ADC \quad (\text{Exterior angle property})$$

$$\Rightarrow 105^\circ = 60^\circ + \angle ADC$$

$$\Rightarrow \angle ADC = 105^\circ - 60^\circ$$

$$\Rightarrow \angle ADC = 45^\circ$$

Hence,  $\angle ADC = 45^\circ$ .

(iii) In  $\triangle ABD$ , side BA is extended to E

$$\angle DAE = \angle ABD + \angle ADB \quad (\text{Exterior angle property})$$

$$\Rightarrow \angle DAE = 40^\circ + 45^\circ \quad (\because \angle ADB = \angle ADC = 45^\circ)$$

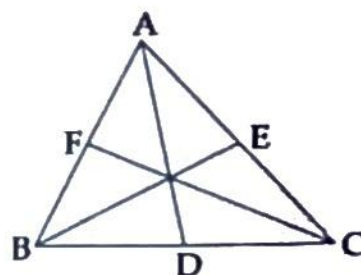
$$\Rightarrow \angle DAE = 85^\circ$$

## MEDIANS OF A TRIANGLE

A line segment joining any vertex of a triangle to the mid-point of its opposite side is called a **median** of the triangle. In other words, a median connects a vertex of a triangle to the mid-point of the opposite side.

In the given  $\triangle ABC$ , D, E and F are the mid-points of the sides BC, CA and AB respectively.

Therefore, AD, BE and CF are the medians of  $\triangle ABC$ .



## ALTITUDE OF A TRIANGLE

The altitude (height) is the shortest distance from the vertex to the opposite side and the shortest distance is the perpendicular distance.

In the figure, the height is given by the line segment that starts from A, comes straight down to BC and is perpendicular to BC.

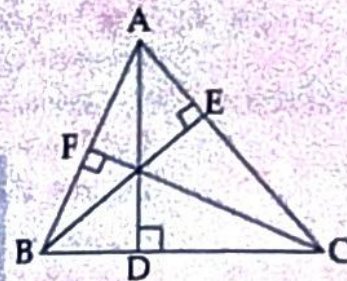
A perpendicular drawn from any vertex to its opposite side is called an **altitude** of the triangle.

Through each vertex of a triangle an altitude can be drawn.



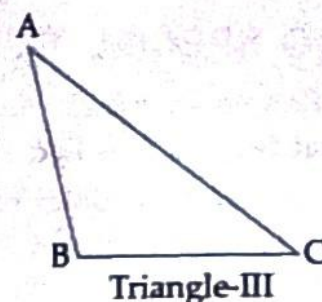
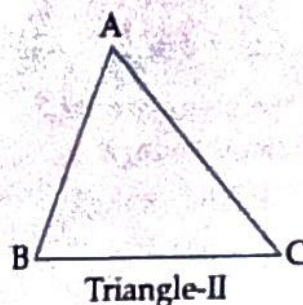
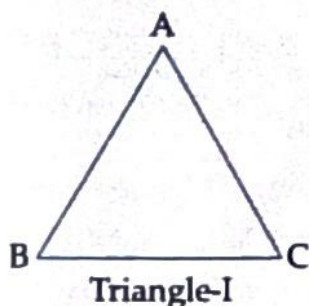
In  $\triangle ABC$ ,  $AD \perp BC$ ,  $BE \perp AC$ , and  $CF \perp AB$ . Therefore,  $AD$ ,  $BE$  and  $CF$  are the altitudes of the triangle  $ABC$ . The maximum number of altitudes that can be drawn in a triangle is three.

**Note :** In a scalene triangle, the altitudes are of different lengths. In an isosceles triangle, altitudes to the equal sides are equal in length. In an equilateral triangle, all three altitudes are equal in length.



## TRIANGLE INEQUALITY PROPERTY

Draw three triangles as shown in the figure below :



Measure the length of each side of the triangles and complete the given table.

| Triangle | Sides of $\triangle ABC$ |    |    | Computations |           |           |
|----------|--------------------------|----|----|--------------|-----------|-----------|
|          | AB                       | BC | CA | $AB + BC$    | $BC + CA$ | $CA + AB$ |
| I        |                          |    |    |              |           |           |
| II       |                          |    |    |              |           |           |
| III      |                          |    |    |              |           |           |

Observe that :

(i)  $AB + BC > CA$

(ii)  $BC + CA > AB$

(iii)  $CA + AB > BC$

Thus, the sum of the lengths of any two sides of a triangle is greater than the length of the third side.

This important property of triangle is known as **triangle inequality property**.

Also, if we measure the three angles of a triangle, we will find that angle opposite to the longest side is the greatest.

**Another important result :** The difference between the lengths of any two sides of a triangle is smaller than the length of the third side.

**Example 1 :** Is it possible to draw triangles with the measurement given below?

(i) 3 cm, 4 cm, 5 cm

(ii) 2 cm, 10 cm, 15 cm

**Solution :** (i) Sides are 3 cm, 4 cm, 5 cm.

Now,  $3 + 4 > 5$ ,  $4 + 5 > 3$ ,  $5 + 3 > 4$

Hence, it is possible to draw a triangle with sides 3 cm, 4 cm and 5 cm.

(ii) Sides are 2 cm, 10 cm and 15 cm.

$2 + 15 > 10$ ,  $2 + 10 > 15$ ,  $10 + 15 > 2$

Since,  $2 + 10 < 15$ , it is not possible to draw a triangle with sides 2 cm, 10 cm and 15 cm.

**Example 2 :** The lengths of two sides of a triangle are 12 cm and 15 cm. Between what two measures should the length of third side fall?

**Solution :** We know that, the sum of the two sides of a triangle is always greater than the third side.

i.e., the third side has to be less than the sum of the two sides.

Therefore, the third side has to be less than  $15 \text{ cm} + 12 \text{ cm} = 27 \text{ cm}$ .

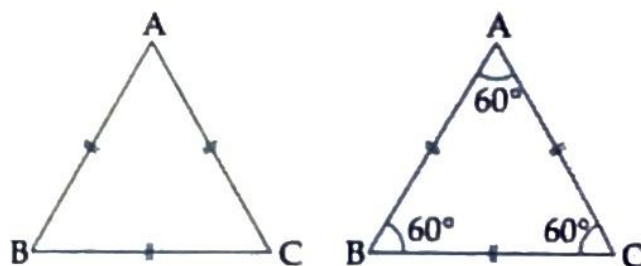
The third side cannot be less than the difference of the two sides. Therefore, the third side has to be more than  $15 \text{ cm} - 12 \text{ cm} = 3 \text{ cm}$ .

Hence, the length of the third side could be any length greater than 3 cm but less than 27 cm.

## TWO SPECIAL TRIANGLES : EQUILATERAL AND ISOSCELES

A triangle in which all the three sides are of equal lengths is called an equilateral triangle.

Let us take two copies of an equilateral triangle ABC as shown in the figure. We keep one of them fixed and place the second triangle on it. It fits exactly onto the first. If we turn the second triangle any way, it still fits on the other exactly.



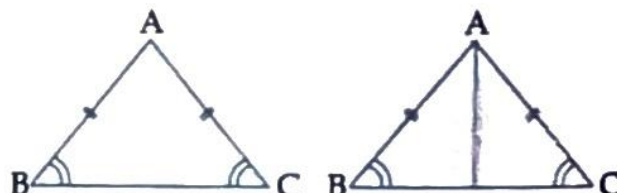
It implies that, if three sides of a triangle are equal, then the three angles are also equal.

We can also define equilateral triangle as a triangle having three equal angles. Since the sum of the angles of a triangle is  $180^\circ$ , each angle of an equilateral triangle measures  $60^\circ$ .

A triangle in which two sides are of equal lengths is called an isosceles triangle.

Consider a triangle ABC.

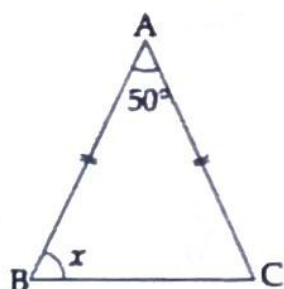
If we fold the triangle ABC such that B falls on C, AB coincides with AC. We find that  $\angle B$  and  $\angle C$  fit on each other exactly.



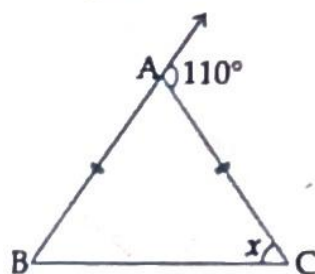
Thus, if two sides of a triangle are equal, then the angles opposite to them are also equal.

**Example 3 :** Find the value of  $x$  in each figure :

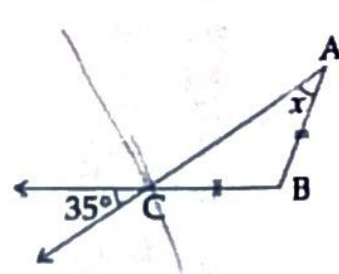
(i)



(ii)



(iii)



**Solution :** (i) In  $\triangle ABC$ ,

$\therefore$

Now,

$\Rightarrow$

$\Rightarrow$

$\Rightarrow$

$\Rightarrow$

$\therefore$

$$AB = AC$$

$$\angle B = \angle C \text{ (Angles opposite to equal sides are equal)}$$

$$\angle A + \angle B + \angle C = 180^\circ \text{ (Angle sum property)}$$

$$50^\circ + 2\angle B = 180^\circ$$

$$2\angle B = 180^\circ - 50^\circ$$

$$2\angle B = 130^\circ \quad (\angle B = \angle C)$$

$$\angle B = 65^\circ$$

$$x = 65^\circ$$

(ii) In  $\triangle ABC$ , side BA is extended.

$$\therefore \text{Exterior } \angle A = \angle B + \angle C$$

$$\Rightarrow 110^\circ = x + x \quad (\because AB = AC, \angle B = \angle C = x)$$

$$\Rightarrow 2x = 110^\circ$$

$$\therefore x = 55^\circ$$

(iii) In  $\triangle ABC$ ,

$$AB = BC$$

$$\therefore \angle A = \angle C \quad (\text{Angles opposite to equal sides are equal})$$

$$x = \angle C$$

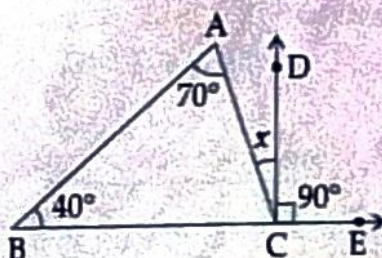
$$\text{But, } \angle C = 35^\circ \quad (\text{vertically opposite angles})$$

$$\therefore x = 35^\circ$$



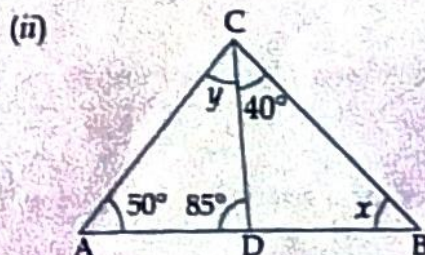
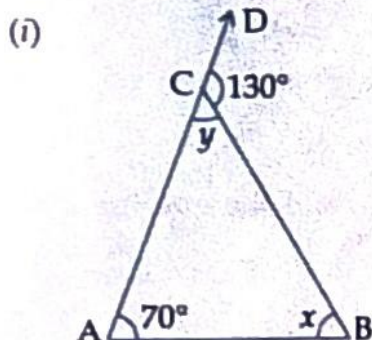
## EXERCISE 12.2

- One side of a triangle is produced and the exterior angle so formed is  $120^\circ$ . If the interior opposite angles are in the ratio 2 : 3, find the measure of each angle of the triangle.
- From the figure, find the value of  $x$  :

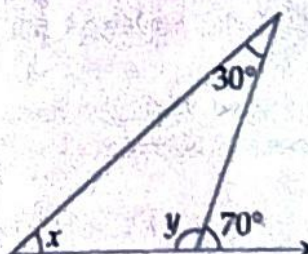


- If the sides of a triangle are produced in order, prove that the sum of the exterior angles so formed is  $360^\circ$ .

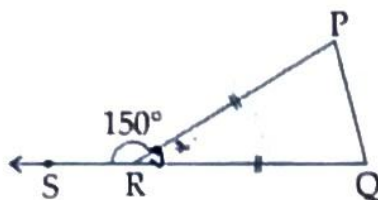
- In the given figures, find the values of  $x$  and  $y$ .



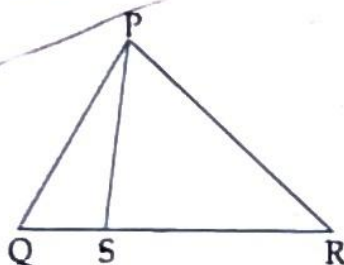
- From the figure given below, find the values of  $x$  and  $y$ .



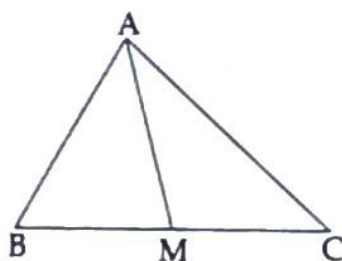
6. An exterior angle of a triangle measures  $120^\circ$  and its interior opposite angles are in the ratio 5 : 7. Find the angles of the triangle.
7. In  $\triangle PRQ$ , side  $QR$  is extended to  $S$ . If  $\angle PRS = 150^\circ$  and  $RP = RQ$ , find the three angles of  $\triangle PRQ$ .



8. The lengths of two sides of a triangle are 6 cm and 8 cm. Between which two numbers can length of the third side fall?
9. Is it possible to have a triangle with the following sides?  
 (i) 6 cm, 3 cm, 2 cm      (ii) 3 cm, 6 cm, 7 cm      (iii) 10.2 cm, 5.8 cm, 4.5 cm
10. In the given figure,  $S$  is a point on the side  $QR$  of  $\triangle PQR$ . Prove that  $PQ + QR + PR > 2 PS$ .



11.  $AM$  is a median of a  $\triangle ABC$ .  
 Prove that  $AB + BC + CA > 2 AM$ .



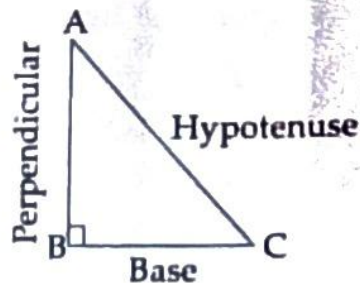
12. Calculate the angles of a triangle  $ABC$  having  $3\angle B = 4\angle C$  and the interior  $\angle A = \frac{3}{7}$  of exterior  $\angle A$ .

## PYTHAGORAS THEOREM

A right angled triangle has one of its angles equals to  $90^\circ$ . The side opposite to the right angle is called the **hypotenuse**. The other two sides that contain right angle are called the **base** and **perpendicular** of the right angled triangle.

A relationship between hypotenuse, base and perpendicular of a **right angled** triangle was given by Greek Mathematician **Pythagoras**, who was born in 570 BCE.

Draw a right angled triangle  $ABC$ , right angled at  $A$ . Let  $AB = c$ ,  $BC = a$  and  $AC = b$ .



Construct a square on each of its sides. Compute the area of these three squares, having sides  $a$ ,  $b$ ,  $c$  units. We find that,

$$a^2 = b^2 + c^2$$

Area of square on the hypotenuse = Area of square on the base + Area of square on the perpendicular

$$(\text{Hypotenuse})^2 = (\text{Base})^2 + (\text{Perpendicular})^2$$

Thus, Pythagoras theorem states that,

"In a right angled triangle, the area of the square on the hypotenuse is equal to the sum of the areas of squares on its remaining two sides."

In other words, "In a right angled triangle, the square of the hypotenuse is equal to the sum of the squares of base and perpendicular."

$$BC^2 = AB^2 + AC^2$$

Conversely, if the Pythagoras property holds, the triangle must be right angled.

**Remark :** The two sides of a right triangle other than the hypotenuse are called its legs.

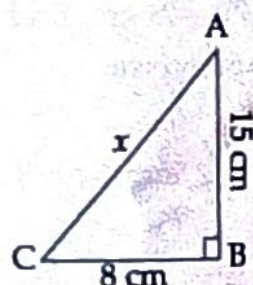
### Pythagorean Triplets

Three positive integers  $a$ ,  $b$ ,  $c$  are said to form a Pythagorean triplet, if  $c^2 = a^2 + b^2$ .

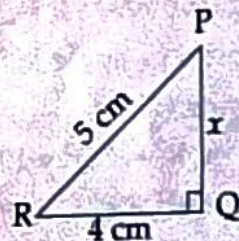
Thus, each of the triplets (3, 4, 5), (5, 12, 13), (7, 24, 25), (8, 15, 17) and (12, 35, 37) is a Pythagorean triplet.

**Example 1 :** Find the unknown length in the following figures :

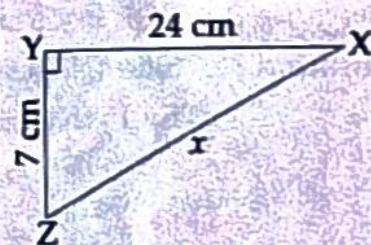
(i)



(ii)



(iii)



**Solution :** (i) We know that the side opposite to the right angle is called the hypotenuse.

(by Pythagoras theorem)

$$\therefore AC^2 = BC^2 + AB^2$$

$$\Rightarrow x^2 = 8^2 + 15^2$$

$$\Rightarrow x^2 = 64 + 225$$

$$\Rightarrow x^2 = 289$$

$$\Rightarrow x^2 = (17)^2$$

$$\Rightarrow x = 17 \text{ cm}$$

(ii) By Pythagoras theorem,

$$PR^2 = QR^2 + PQ^2$$

$$\therefore 5^2 = 4^2 + x^2$$

$$\Rightarrow 25 = 16 + x^2$$

$$\Rightarrow x^2 = 25 - 16 = 9$$

$$\therefore x = 3 \text{ cm}$$

(iii) By Pythagoras theorem,

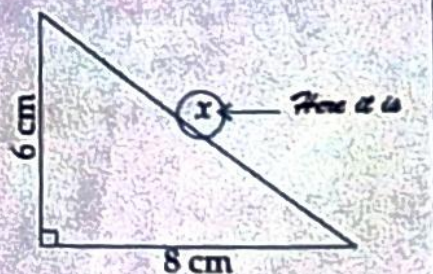
$$XZ^2 = YZ^2 + XY^2$$



### Laughter Time

Q : Find  $x$ .

Ans :



$$\begin{aligned}
 &\Rightarrow x^2 = 7^2 + 24^2 \\
 &\Rightarrow x^2 = 49 + 576 \\
 &\Rightarrow x^2 = 625 \\
 &\Rightarrow x^2 = (25)^2 \\
 &\Rightarrow x = 25 \text{ cm}
 \end{aligned}$$

**Example 2 :** The lengths of the sides of two triangles are given below. Which of them is right triangle?

- (i) 6 cm, 8 cm, 10 cm                      (ii) 8 cm, 5 cm, 10 cm

**Solution :** (i) Three sides are 6 cm, 8 cm, 10 cm.

The longest side is 10 cm.

$$\begin{aligned}
 \text{We have, } 6^2 + 8^2 &= 36 + 64 \\
 &= 100 = 10^2
 \end{aligned}$$

$$\therefore 6^2 + 8^2 = 10^2$$

Hence, the triangle is right angled.

- (ii) Three sides are 8 cm, 5 cm, 10 cm.

The longest side is 10 cm.

$$\begin{aligned}
 \text{We have, } 8^2 + 5^2 &= 64 + 25 \\
 &= 89 \neq 100 = (10)^2
 \end{aligned}$$

$$\therefore 8^2 + 5^2 \neq 10^2$$

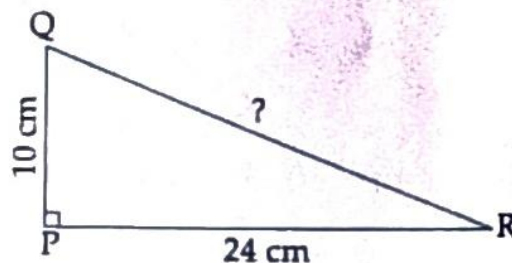
Hence, the triangle is not right angled.

**Example 3 :** PQR is a triangle with  $\angle P = 90^\circ$ . If PQ = 10 cm, PR = 24 cm, find the length of QR.

**Solution :** Side QR being opposite to angle  $90^\circ$  is the hypotenuse.

By Pythagoras theorem,

$$\begin{aligned}
 QR^2 &= PQ^2 + PR^2 \\
 \Rightarrow QR^2 &= (10)^2 + (24)^2 \\
 \Rightarrow QR^2 &= (100 + 576) \\
 \Rightarrow QR^2 &= 676 \\
 \Rightarrow QR^2 &= (26)^2 \\
 \Rightarrow QR &= 26
 \end{aligned}$$



Hence, QR measures 26 cm.

**Example 4 :** A tree is broken at a height of 5 m from the ground and its top touches the ground at a distance of 12 m from the base of the tree. Find the original height of the tree.

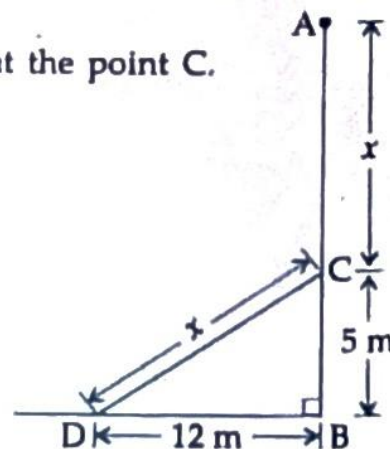
**Solution :** Let AB be the tree of height  $(x + 5)$  m before it broke at the point C.

Let the top A touch the ground at D after it broke.

Then  $\triangle DBC$  is a right triangle.

From the figure,

$$\begin{aligned}
 CD^2 &= DB^2 + BC^2 \\
 x^2 &= (12^2 + 5^2) \\
 \Rightarrow x^2 &= (144 + 25) \\
 \Rightarrow x^2 &= 169
 \end{aligned}$$



$$\Rightarrow x^2 = (13)^2$$

$$\Rightarrow x = 13$$

Original height of the tree =  $x + 5 = 13 + 5 = 18$  m

Hence, the original height of the tree was 18 m.

**Example 5 :** Find the length of the diagonal of a rectangle having sides 16 cm and 12 cm.

**Solution :** Let ABCD be a rectangle.

AD = 16 cm, CD = 12 cm

Diagonal AC is the hypotenuse of right angled  $\triangle ADC$  right angled at D.

$\therefore$  By Pythagoras theorem,

$$AC^2 = AD^2 + CD^2$$

$$\Rightarrow AC^2 = (16^2 + 12^2)$$

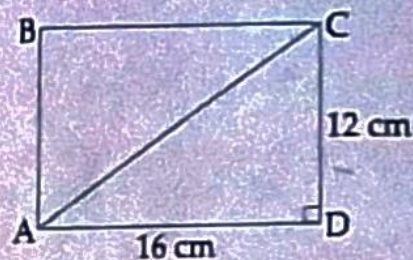
$$\Rightarrow AC^2 = (256 + 144)$$

$$\Rightarrow AC^2 = 400$$

$$\Rightarrow AC^2 = (20)^2$$

$$\Rightarrow AC = 20$$

Hence, length of the diagonal is 20 cm.



**Example 6 :** Two poles, 13 m and 18 m high stand upright on a playground. If their feet are 12 m apart, find the distance between their tops.

**Solution :** Let AB and CD be two poles of heights 13 m and 18 m respectively. Draw  $AE \parallel BD$ .

Then,

$$CE = DC - DE$$

$$= (18 - 13) \text{ m} = 5 \text{ m}$$

In right angled  $\triangle AEC$ ,

$$AE = BD = 12 \text{ m}, CE = 5 \text{ m}$$

By Pythagoras theorem,

$$AC^2 = AE^2 + CE^2$$

$$\Rightarrow AC^2 = (12^2 + 5^2)$$

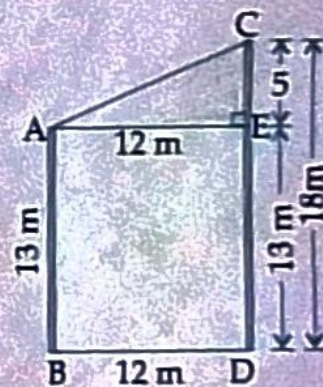
$$\Rightarrow AC^2 = (144 + 25)$$

$$\Rightarrow AC^2 = 169$$

$$\Rightarrow AC^2 = (13)^2$$

$$\therefore AC = 13$$

Hence, the distance between the tops of the poles is 13 m.



**Example 7 :** The diagonals of a rhombus are 30 cm and 16 cm. Find its perimeter.

**Solution :** We know that diagonals of a rhombus bisect each other at right angles.

Let ABCD be a rhombus and AC and BD be its two diagonals,

where AC = 30 cm and BD = 16 cm.

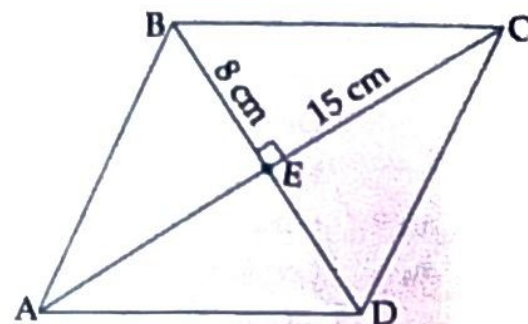
Let the two diagonals bisect at E.

Then,  $\angle BEC = 90^\circ$  and EC = 15 cm and BE = 8 cm

$$\left( \because EC = \frac{1}{2} AC \text{ and } BE = \frac{1}{2} BD \right)$$

In right angled  $\triangle BEC$ ,

$$\begin{aligned} BC^2 &= BE^2 + EC^2 \\ \Rightarrow BC^2 &= (8^2 + 15^2) \\ \Rightarrow BC^2 &= (64 + 225) \\ \Rightarrow BC^2 &= 289 \\ \Rightarrow BC^2 &= (17)^2 \\ \therefore BC &= 17 \end{aligned}$$



In a rhombus, all sides are of equal lengths.

$$\begin{aligned} \therefore \text{Perimeter of rhombus} &= 4 \times \text{side} \\ &= (4 \times 17) \text{ cm} \\ &= 68 \text{ cm} \end{aligned}$$

Hence, the perimeter of rhombus is 68 cm.

**Example 8 :** A ladder 13 m long reaches a window which is 5 m above the ground, on one side of the street. Keeping its foot at the same point, the ladder is turned to the other side of the street to reach a window at a height of 12 m. Find the width of the street.

**Solution :** Let PQ be the street and R be the foot of the ladder.

Let S and T be the windows at the heights of 5 m and 12 m respectively from the ground.

Then, RS and RT are two positions of the ladder.

Clearly PS = 5 m, QT = 12 m, RT = RS = 13 m

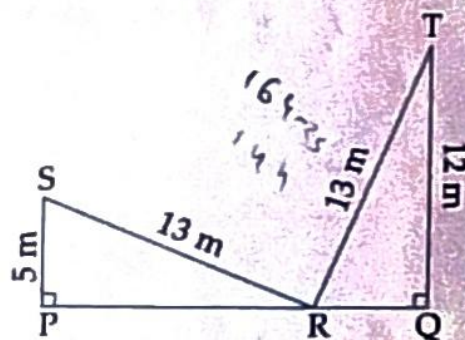
From right  $\triangle SPR$ , we have

$$\begin{aligned} PR^2 + PS^2 &= RS^2 \\ \Rightarrow PR^2 &= RS^2 - PS^2 \\ \Rightarrow PR^2 &= (13^2 - 5^2) \\ \Rightarrow PR^2 &= (169 - 25) \\ \Rightarrow PR^2 &= 144 \\ \Rightarrow PR^2 &= (12)^2 \\ \Rightarrow PR &= 12 \end{aligned}$$

From right  $\triangle RQT$

$$\begin{aligned} RQ^2 + QT^2 &= RT^2 \\ \Rightarrow RQ^2 &= RT^2 - QT^2 \\ \Rightarrow RQ^2 &= (13^2 - 12^2) \\ \Rightarrow RQ^2 &= (169 - 144) \\ \Rightarrow RQ^2 &= 25 \\ \Rightarrow RQ^2 &= (5)^2 \\ \Rightarrow RQ &= 5 \end{aligned}$$

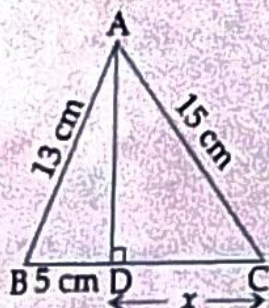
Hence, width of the street = PQ = PR + RQ = (12 + 5) m = 17 m.





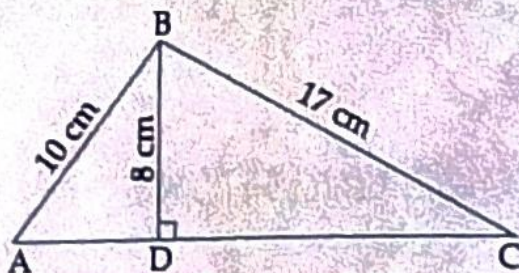
## EXERCISE 12.3

1. ABC is a triangle, right angled at C. If  $AB = 25$  cm and  $AC = 7$  cm, find BC.
2. In the adjoining figure, find the value of  $x$ .



$$\begin{array}{r} 15 \\ 5 \overline{) 75} \\ \underline{50} \phantom{0} \\ 250 \\ \underline{250} \\ 0 \end{array}$$

3. A 10 m long ladder reaches a window 8 m high from the ground on placing it against a wall at a distance  $x$ . Find the distance ' $x$ '.
4. Identify the 'Pythagorean triplets' from the following :  
(i) 3, 7, 9                      (ii) 7, 24, 25                      (iii) 4, 5, 7
5. The hypotenuse of a right angled triangle is 25 cm. If one of the remaining two sides is 15 cm, find the length of its third side.
6. A man goes 10 m due east and then 24 m due north. How far is he from the starting point?
7. The sides of a triangle are 15 cm, 36 cm and 39 cm. Verify that it is a right angled triangle.
8. A ladder 4.5 m long is placed against a wall in such a way that the foot of the ladder is 2.7 m away from the wall. Find the height of the wall to which the ladder reaches.
9. Two poles of heights 9 m and 14 m stand upright on a plane ground. If the distance between their feet is 12 m, find the distance between their tops.
10. The sides of a triangle are of lengths 6.5 cm, 6 cm and 2.5 cm. Is this triangle a right triangle? If yes, which is the hypotenuse?
11. In the given figure, find the length AC.



12. The adjacent sides of a rectangle are 3.6 cm and 1.5 cm. Find the length of the diagonal.
13. ABC is an isosceles right triangle, right angled at C.  
Prove that :  $AB^2 = 2 AC^2$ .



## FACTS TO REMEMBER

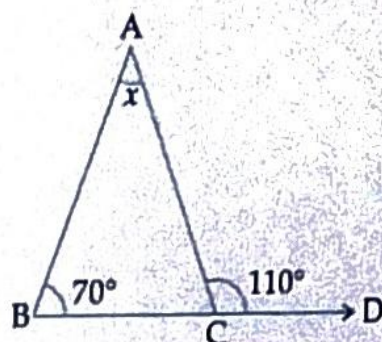
- If A, B, C are three non-collinear points, then the closed figure formed by three line segments AB, BC and CA is called a triangle with vertices A, B and C.
- A triangle is said to be
  - (i) an equilateral triangle if all of its sides are equal;
  - (ii) an isosceles triangle if two of its sides are equal;
  - (iii) a scalene triangle if all of its sides are of different lengths.
- A triangle is said to be
  - (i) an acute triangle if each of its angles measures less than  $90^\circ$ .
  - (ii) a right triangle if one of its angles measures  $90^\circ$ .
  - (iii) an obtuse triangle if one of its angles measures more than  $90^\circ$ .
- The sum of the angles of a triangle is  $180^\circ$ .
- A line segment joining a vertex of the triangle to the mid-point of its opposite side is called a median of the triangle. A triangle has 3 medians.
- A perpendicular drawn from a vertex of a triangle to its opposite side is called an altitude of the triangle. A triangle has 3 altitudes.
- The measure of any exterior angle of a triangle is equal to the sum of the measures of its interior opposite angles.
- The sum of the lengths of any two sides of a triangle is greater than the length of the third side.
- In a right angled triangle, the side opposite to  $90^\circ$  is called the hypotenuse and the other two sides are called its legs (perpendicular and base).
- In a right angled triangle, the square of the hypotenuse is equal to the sum of the squares of the remaining two sides. This is known as Pythagoras theorem.



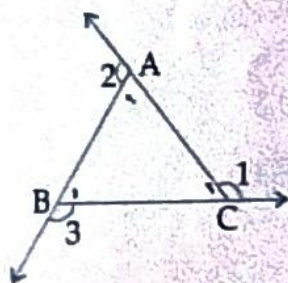
## MULTIPLE CHOICE QUESTIONS

Choose the correct alternatives in each of the following :

1. Which of the following is not a Pythagorean triplet?  
(a) 5, 12, 13  
(b) 10, 15, 25  
(c) 15, 36, 39  
(d) 10, 24, 26
2. In the given figure, side BC of  $\triangle ABC$  is extended to D. If  $\angle ACD = 110^\circ$  and  $\angle ABC = 70^\circ$ , then measure of x is

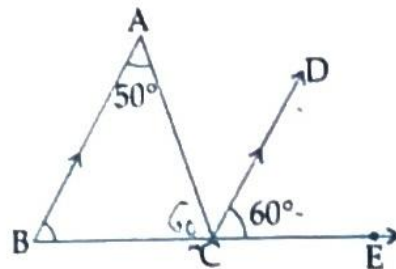


- (a)  $40^\circ$                       (b)  $50^\circ$                       (c)  $60^\circ$                       (d)  $55^\circ$
3. In a right angled triangle, right angled at B, if  $AB = 5$  cm and  $AC = 13$  cm, then BC is equal to  
 (a) 8 cm                      (b) 12 cm                      (c) 10 cm                      (d) 5 cm
4. The length of a rectangle is 12 m and the length of its diagonal is 15 m. The width of the rectangle is  
 (a) 8 m                      (b) 10 m                      (c) 11 m                      (d) 9 m
5. Which of the following cannot be the lengths of the sides of a triangle?  
 (a) 3 cm, 4 cm, 5 cm  
 (b) 5 cm, 6 cm, 11 cm  
 (c) 7 m, 8 m, 14 m  
 (d) 3 cm, 3 cm, 3 cm
6. The sides of a triangle are produced in order. Then, the sum of the exterior angles,  $\angle 1 + \angle 2 + \angle 3$  is

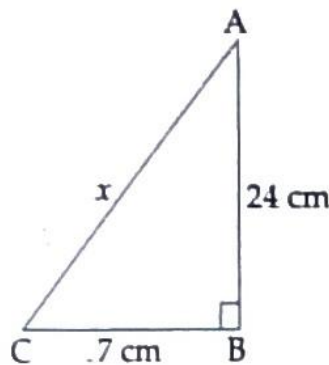


- (a)  $180^\circ$                       (b)  $360^\circ$   
 (c)  $540^\circ$                       (d) none of these
7. Two angles of a triangle are equal and third angle measures  $80^\circ$ . The measure of each of the equal angles is  
 (a)  $55^\circ$                       (b)  $60^\circ$                       (c)  $50^\circ$                       (d)  $65^\circ$
8. The square of the hypotenuse of a right triangle is 50 cm. If the two legs of right angled triangle are equal, the length of each leg is  
 (a) 10 cm                      (b) 5 cm  
 (c) 12.5 cm                      (d) none of these
9. If the sides of a rectangle are 16 cm and 12 cm, the length of each of its diagonals is  
 (a) 20 cm                      (b) 24 cm  
 (c) 18 cm                      (d) none of these

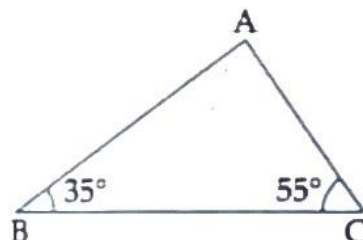
10. In the given  $\triangle ABC$ ,  $\angle A = 50^\circ$ ,  $CD \parallel BA$ ,  $\angle ECD = 60^\circ$ , then  $\angle ACB$  is



- (a)  $50^\circ$  (b)  $60^\circ$  (c)  $70^\circ$  (d)  $80^\circ$
11. In the adjacent figure,  $\triangle ABC$  is right angled at B,  $AB = 24$  cm and  $BC = 7$  cm. The value of  $x$  is



- (a) 28 cm (b) 27 cm (c) 26 cm (d) 25 cm
12. In the given  $\triangle ABC$ ,  $\angle B = 35^\circ$ ,  $\angle C = 55^\circ$ , then



- (a)  $AB^2 = BC^2 + AC^2$  (b)  $BC^2 = AB^2 + AC^2$   
 (c)  $AC^2 = AB^2 + BC^2$  (d) none of these



## MENTAL MATHS CORNER

Mark the following statements as 'True' or 'False'.

1. A triangle with sides 2.5 cm, 2 cm and 1.5 cm is possible.
2. If AM is a median of  $\triangle ABC$ , then  $AB + BC + CA > 2 AM$ .
3. We can have a triangle with two right angles.
4. If two angles of a triangle are  $30^\circ$  and  $70^\circ$ , then the third angle is  $90^\circ$ .
5. In an isosceles triangle, two angles are equal.
6. The sum of the lengths of two sides of a triangle is always less than the third side.

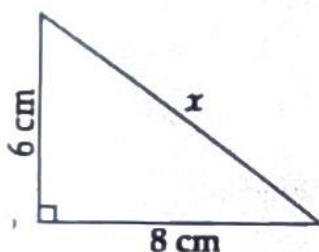
7. If  $O$  is any point in the interior of a  $\triangle ABC$ , then  $OA + OC > AC$ .
8. In a  $\triangle ABC$ , if  $\angle A = 110^\circ$ ,  $\angle B = 40^\circ$ , then the longest side is  $BC$  and the smallest side is  $AB$ .
9. A triangle can be drawn with sides 24 cm, 1.6 cm and 4 cm.
10. Three numbers  $a$ ,  $b$  and  $c$  form a Pythagorean triplet, if  $a^2 + b^2 - c^2 = 0$ .
11. A triangle with two acute angles is not possible.
12. Of all the line segments that can be drawn to a given line from a given point lying outside it, the perpendicular is the shortest.



## REVIEW EXERCISE

1. Find the unknown length  $x$  in the following figures :

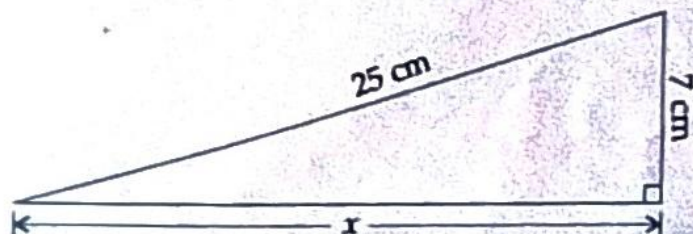
(i)



(ii)

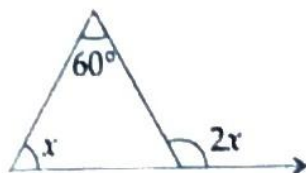


(iii)



2. The angles of a triangle are in the ratio 1 : 2 : 1. Find all the angles of the triangle. Name the triangle in two different ways.
3. Each of the two equal angles of an isosceles triangle is twice the third angle. Find the angles of the triangle.
4. Is it possible to draw a triangle, the lengths of whose sides are given below?
  - (i) 1 cm, 1 cm, 1 cm
  - (ii) 6 cm, 7 cm, 14 cm
  - (iii) 5 cm, 7 cm, 12 cm
  - (iv) 2 cm, 10 cm, 15 cm
5. A ladder 37 m long reaches a window which is 12 m above the ground, on one side of the street. Keeping its foot at the same point, the ladder is turned to the other side of the street to reach a window at a height of 35 m. Find the width of the street.
6. Find the length of the diagonal of a rectangle whose sides are 15 cm and 8 cm.
7. A 5 m long ladder when set against the wall of a house reaches to a height of 4.8 m. How far is the foot of the ladder from the wall?
8. Find the measures of all the angles of an isosceles right-angled triangle.

9. Find the value of  $x$ , from the adjoining figure.

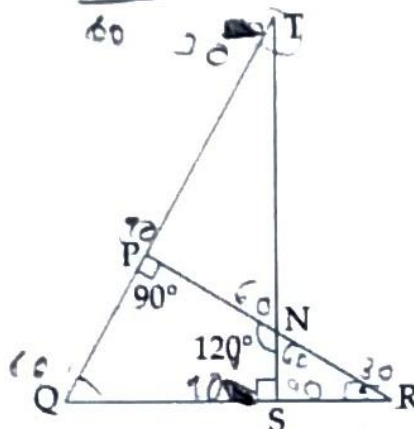


10. PQR is a right triangle, right angled at P. QP is extended to T and  $TS \perp QR$  intersecting PR at N. If  $\angle PNS = 120^\circ$ , find :

(i)  $\angle QTS$   
30

(ii)  $\angle PQR$   
60

(iii)  $\angle QRP$   
30



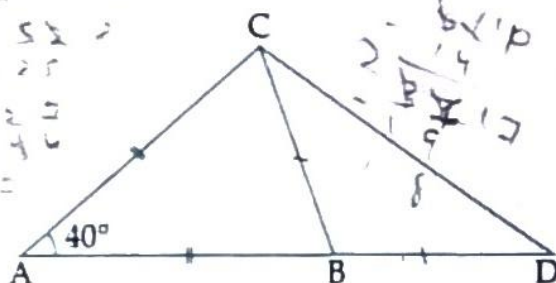
Handwritten calculations for question 10:  
 $90^\circ - 60^\circ = 30^\circ$   
 $180^\circ - 120^\circ = 60^\circ$   
 $90^\circ - 60^\circ = 30^\circ$

11. In the given figure,  $\angle CAB = 40^\circ$ ,  $AC = AB$  and  $BC = BD$ . Find :

(i)  $\angle ACB$

(ii)  $\angle CDB$

Handwritten calculations for question 11:  
 $180^\circ - 40^\circ = 140^\circ$   
 $140^\circ / 2 = 70^\circ$



Handwritten calculations for question 11:  
 $180^\circ - 40^\circ = 140^\circ$   
 $140^\circ / 2 = 70^\circ$   
 $180^\circ - 70^\circ - 70^\circ = 40^\circ$

12. A ladder 25 m long, when set against the wall of a house just reaches a window at a height of 24 m. How far is the lower end of the ladder from the base of the wall?

## HOTS QUESTIONS

- The measure of one of the exterior angles of a triangle is twice of one of its interior opposite angles and measure of other opposite interior angle is  $60^\circ$ . Name the type of the triangle justifying your answer.
- A vertical tree broke at a point and top point of upper portion touches the ground at a distance of 6 m from its base. If the point where it broke is at a height of 2.5 metre from the ground, find the total height of the tree before it broke.
- (i)  $\triangle ABC$  is an equilateral triangle. From A, an altitude AD is drawn. Find the measure of  $\angle BAD$ .  
 (ii) In an isosceles  $\triangle PQR$ ,  $PQ = PR$  and PT is perpendicular to QR. If  $QT = 3$  cm, what will be the length of TR?



## ANSWERS

### Exercise 12.1

1. (i) Yes (ii) No (iii) Yes (iv) No 2.  $45^\circ$
3.  $36^\circ, 72^\circ, 72^\circ$  4.  $48^\circ$  5.  $60^\circ$
6. (i)  $50^\circ$  (ii)  $40^\circ$  (iii)  $50^\circ$  7.  $45^\circ$  8.  $540^\circ$
9.  $75^\circ, 60^\circ, 45^\circ$
10.  $\angle A = 90^\circ, \angle B = 60^\circ, \angle C = 30^\circ$
11.  $51^\circ$

### Exercise 12.2

1.  $48^\circ, 72^\circ, 60^\circ$  2.  $20^\circ$
4. (i)  $x = 60^\circ, y = 50^\circ$  (ii)  $x = 45^\circ, y = 45^\circ$
5.  $x = 40^\circ, y = 110^\circ$  6.  $50^\circ, 70^\circ, 60^\circ$
7.  $30^\circ, 75^\circ, 75^\circ$
8. Less than 14 cm but greater than 2 cm.
9. (i) No (ii) Yes (iii) Yes
12.  $\angle A = 54^\circ, \angle B = 72^\circ, \angle C = 54^\circ$

### Exercise 12.3

1. 24 cm 2. 9 cm 3. 6 m 4. (ii)
5. 20 cm 6. 26 m
7. Yes, it is right angled triangle
8. 3.6 m 9. 13 m 10. Yes, 6.5 cm
11. 21 cm 12. 3.9 cm

### Multiple Choice Questions

1. (b) 2. (a) 3. (b) 4. (d) 5. (b)
6. (b) 7. (c) 8. (b) 9. (a) 10. (c)
11. (d) 12. (b)

### Mental Maths Corner

1. True 2. True 3. False
4. False 5. True 6. False
7. True 8. True 9. False
10. True 11. False 12. True

### Review Exercise

1. (i) 10 cm (ii) 10 cm (iii) 24 cm
2.  $45^\circ, 90^\circ, 45^\circ$ ; Isosceles triangle, right angled triangle
3.  $72^\circ, 72^\circ, 36^\circ$
4. (i) Yes (ii) No (iii) No (iv) No
5. 47 m 6. 17 cm 7. 14 m
8.  $45^\circ, 90^\circ, 45^\circ$  9.  $60^\circ$
10. (i)  $30^\circ$  (ii)  $60^\circ$  (iii)  $30^\circ$
11. (i)  $70^\circ$  (ii)  $35^\circ$  12. 7 m

### HOTS Questions

1. Equilateral triangle 2. 9 m
3. (i)  $30^\circ$  (ii) 3 cm

# MATHS LAB ACTIVITY

**Objective :** To verify Pythagoras theorem by paper cutting and arranging.

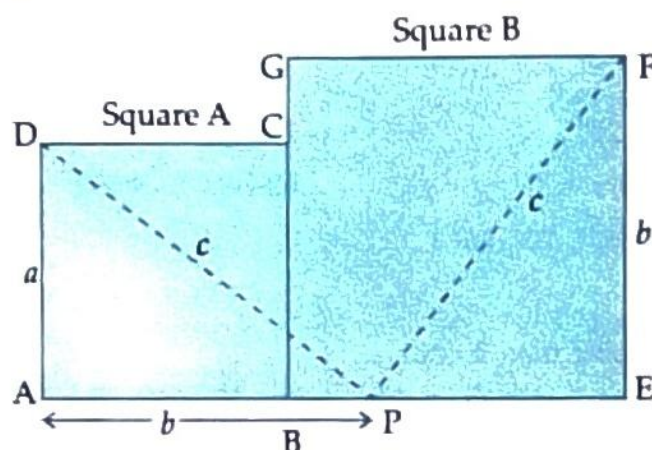
**Activity :**

**Step 1 :** Construct two squares ABCD and BEFG of sides  $a$  and  $b$  units in a row.

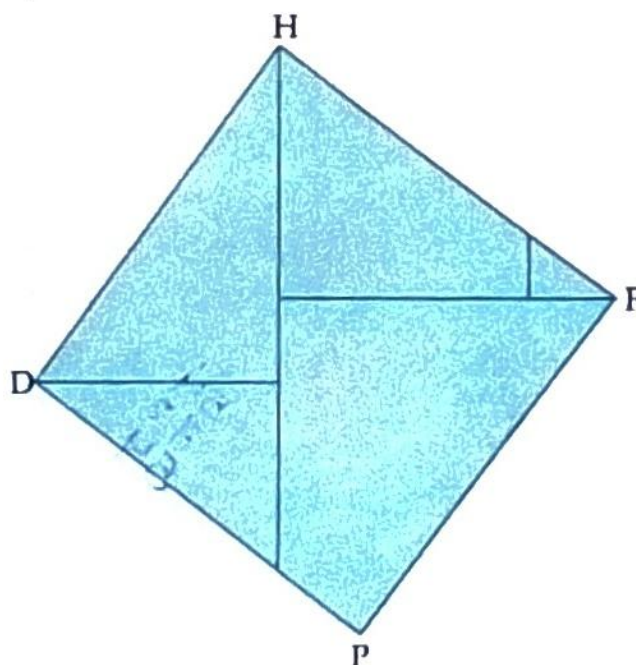
**Step 2 :** Take a point P on AE such that  $AP = b$  units

Then,  $PE = (a + b) - b = a$  units

**Step 3 :** Join PD and PF. Then  $\triangle DAP$  and  $\triangle PEF$  are right angled triangles with sides  $a$ ,  $b$  and hypotenuse  $c$ .



**Step 4 :** Cut off triangles DAP and PEF and paste these as shown in the figure to obtain a square DPEH of side  $c$ .



Area of square DPEH = Area of square ABCD + Area of square BEFG

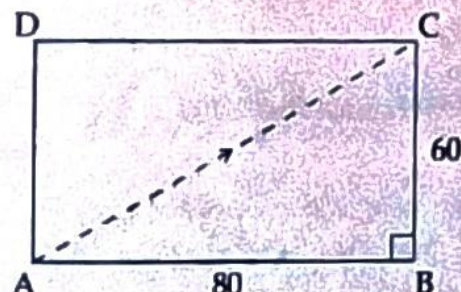
$$c^2 = a^2 + b^2$$

Verify the Pythagoras theorem, for a right triangle having sides 8 cm, 6 cm and hypotenuse 10 cm.

## NCERT Exemplar Problems

**Example 1 :** Jayanti takes shortest route to her home by walking diagonally across a rectangular park. The park measures 60 metres  $\times$  80 metres. How much shorter is the route across the park than the route around its edges?

**Solution :** Park is rectangular in shape with dimensions : length = 60 m and breadth = 80 m. Let ABCD be the park with her home situated at C.



Now, since  $\Delta ABC$  is a right angled triangle

$$\therefore AC^2 = AB^2 + BC^2 \quad [\text{Using Pythagoras theorem}]$$

$$\Rightarrow AC^2 = (80)^2 + (60)^2$$

$$\Rightarrow AC^2 = 6400 + 3600 \Rightarrow AC^2 = 10000 \Rightarrow AC = 100 \text{ m}$$

Reaching C through B Jayanti has to travel  $80 \text{ m} + 60 \text{ m} = 140 \text{ m}$

Her route diagonally is shorter by  $(140 - 100) \text{ m} = 40 \text{ m}$

**Example 2 :** The foot of a ladder is 6 m away from its wall and its top reaches a window 8 m above the ground. (a) Find the length of the ladder. (b) If the ladder is shifted in such a way that its foot is 8 m away from the wall, to what height does its top reach?

**Solution :**  $\Delta ABC$  is right angled triangle.

(a) Using Pythagoras theorem

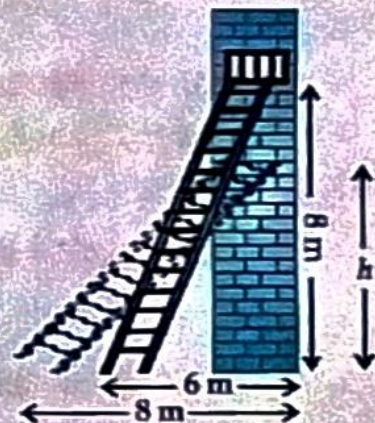
$$\Rightarrow AC^2 = AB^2 + BC^2$$

$$\Rightarrow AC^2 = (8)^2 + (6)^2$$

$$\Rightarrow AC^2 = 64 + 36$$

$$\Rightarrow AC^2 = 100 \Rightarrow AC = 10 \text{ m.}$$

$\therefore$  Length of the ladder = 10 m



(b) After shifting, let the new position of ladder be DE.

Now,  $\Delta DBE$  is also right angled triangle

$$\therefore DE^2 = DB^2 + BE^2 \quad [\text{Using Pythagoras theorem}]$$

$$\Rightarrow 100 = h^2 + 64 \quad [\because DE = AC = 10]$$

$$\Rightarrow h^2 = 100 - 64 \Rightarrow h^2 = 36$$

$$\Rightarrow h = 6 \text{ m}$$

Hence, after shifting the ladder 8 m from the wall, it will reach up to 6 m height.

